

Chat filter: accuracy defect

The filter matches **raw substrings**, so ordinary words containing a blocked fragment are censored — **cucumber** is blocked because of three letters inside it — while the same design lets deliberate evasion straight through. It is simultaneously too aggressive and too weak, and both symptoms have one cause.

VERIFIED LIVE, IN GAME

TYPED	RESULT	WHY
cucumber	blocked	3-letter fragment
Heisenberg	blocked	fragment
thank	was blocked	since fixed
okkk	blocked	contains kkk
okay	fine	control

okkk blocking while okay passes proves matching is **substring-based**, not word-based. These came up in ordinary conversation, not by hunting for edge cases.

WHAT THAT COSTS, AT SCALE

The filter list shipped in the client, run against the game's own English item and skill names:

Freezing → ****zing

Immunity Control → i**unity control

Savage Card → **vage card

Card Skills → card ski**s

Downpour → downpo** Ignite → igni**

46.1%

ITEM & SKILL NAMES

69.0%

ALL LOCALIZED STRINGS

37.1%

ENGLISH DICTIONARY

WORSE OUTSIDE ENGLISH

LOCALE	OWN NAMES FLAGGED
Portuguese	52.1%
Indonesian	48.0%
English	46.1%
Thai	30.0%
Vietnamese	9.7%
Chinese (Trad)	9.7%
Chinese (Simp)	8.4%
Korean	0.0%

The list damages the newer markets roughly **five times more** than its home market. Portuguese, a launch market, is worst hit.

ROOT CAUSE

The list is **95,660 entries, 93.4% Chinese** — market-compliance material (political terms, regional gambling domains, precursor chemicals), appropriate for the market it was written for. Rebirth launched in China in Dec 2025 and globally in Jul 2026; the list travelled with the build.

11,872 entries are two characters long; 497 are a single character. As substrings they sit inside thousands of ordinary words: *te* in *ignite*, *ll* in *skills*, *ur* in *curse*, 日 in 日光. Chinese suffers the identical defect — 日 fires 438 times inside the game's own Chinese names — so a fix helps that build too.

NOT A REQUEST TO FILTER LESS

Against ten common obfuscation strategies the list catches **22.9%**. Spaced-out text, full-width characters and homoglyphs pass untouched. Of 25 common English profanity terms tested, **6** are present at all.

So it blocks cucumber and passes f u c k. Fixing this makes the filter **more accurate in both directions**, not more permissive.

WHAT WOULD FIX IT

- Run the self-test.** ~40 lines, no dependencies, nothing to adopt. Enforces one invariant: *no string we authored may be flagged by our own filter*. Has a ratchet, so it goes into CI today.
- Rank your own entries by damage.** False positives are extremely concentrated — on the copy measured, **20 entries caused 89%** of them, and only 191 of 5,695 needed any change. Emits a CSV with per-entry blast radius and a recommended action: *scope to locale* (deletes nothing), *whole-word only*, or *delete*.
- Fix the matching.** Word boundaries plus a rescue allowlist. Same vocabulary, no new words: dictionary false positives **37.1%** → **0.0%**; with a lexicon generated from your own localization, own-content false positives **46.1%** → **0.0%** while obfuscation recall rises **22.9%** → **84.3%**.

INPUT	TODAY	WITH THE FIX
cucumber	blocked	allowed
Heisenberg	blocked	allowed
okkk	blocked	still blocked
kkk	blocked	still blocked
f u c k	passes	caught

TWO THINGS WORTH KNOWING

Nothing here needs a client update. The filter check is already a server round trip — the client sends candidate text and receives a masked string back. All of the above changes only the server side of that call. No client build, no store review.

59.6% of the list cannot affect any outcome, because a shorter entry already matches inside each of those 57,051 entries. Substring matching has collapsed a 95,660-entry list into the behaviour of its shortest ~38,000. Word-boundary matching does not weaken the list — it restores the other 57,051 to usefulness.

One caveat, stated plainly. The list analysed is the copy shipped in the client, and it does not fully match what the server enforces: *cucumber* is blocked live, yet the responsible fragment is absent from the client copy. **So please don't action any specific row we name** — run the tools against your own list. The class of defect is proven by the live observations above and holds regardless; only the exact rows are yours to generate.